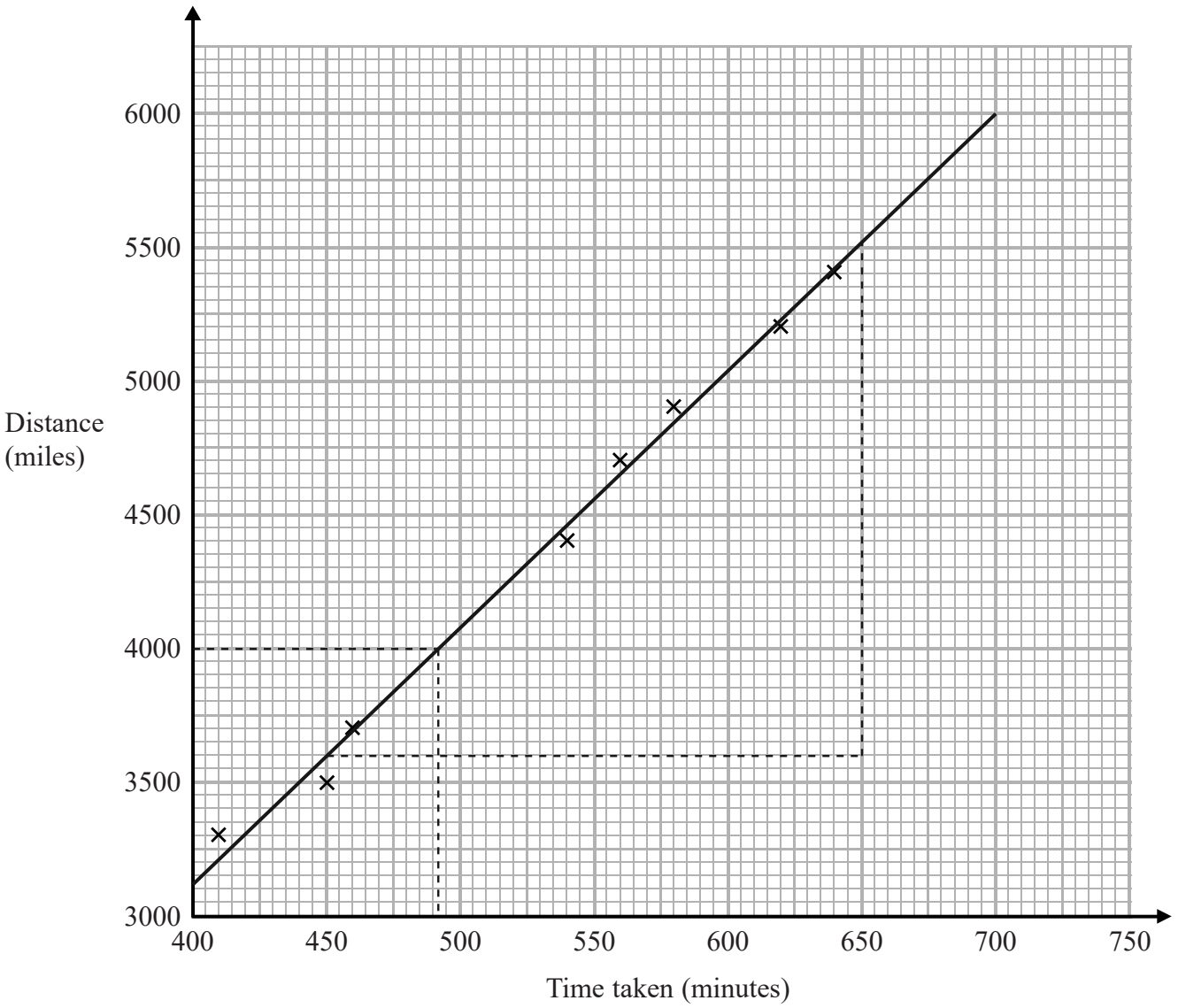
You must write down all the stages in your working.

1 Oliver records the distance from London to each of eight cities in the USA.
He also records the time taken to fly from London to each of these cities.

The scatter graph shows this information.



Gradient triangle:200min across,1920 miles up.

1 mark
Balanced best-fit line

Chicago is a city in the USA.
Chicago is 4000 miles from London.

(a) (i) By drawing a line of best fit, find an estimate for the time taken to fly from London to Chicago.

4000 miles → about 492 minutes.

1 mark
Read best-fit line

..... minutes

(2)

(ii) Why is your answer to part (i) only an estimate?

1 mark
Only a prediction

The points are scattered; a best-fit line can vary.

(1)

(b) (i) Calculate the gradient of your line of best fit.

$$\frac{5520 - 3600}{650 - 450} = 1920 \div 200 = 9.6$$

1 mark
Rise ÷ run

Gradient = 9.6 miles per minute.

1 mark
9.6

(2)

(ii) Give an interpretation of the gradient of your line of best fit.

1 mark
Speed in miles/min

Estimated speed: 9.6 miles travelled per extra minute.

(1)

(Total for Question 1 is 6 marks)